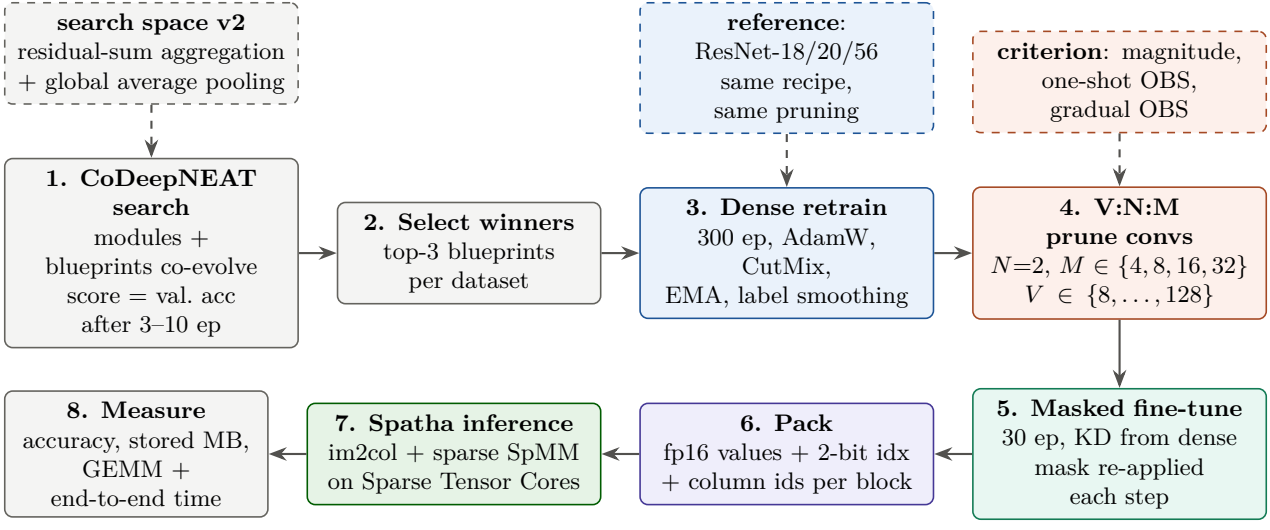

search space v2
residual-sum aggregation
+ global average pooling
reference:
ResNet-18/20/56
same recipe,
same pruning
criterion: magnitude,
one-shot OBS,
gradual OBS
1. CoDeepNEAT search
modules +
blueprints co-evolve
score = val. acc
after 3–10 ep
2. Select winners
top-3 blueprints
per dataset
3. Dense retrain
300 ep, AdamW,
CutMix,
EMA, label smoothing
4. V:N:M prune convs
N=2, M ∈ {4, 8, 16, 32}
V ∈ {8, . . . , 128}
5. Masked fine-tune
30 ep, KD from dense
mask re-applied
each step
6. Pack
fp16 values + 2-bit idx
+ column ids per block
7. Spatha inference
im2col + sparse SpMM
on Sparse Tensor Cores
8. Measure
accuracy, stored MB,
GEMM +
end-to-end time